

Chapter 7: Constitutive Relations

Reading: [Fung/94] Chapter 7

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Contents

(§7.1) Definition

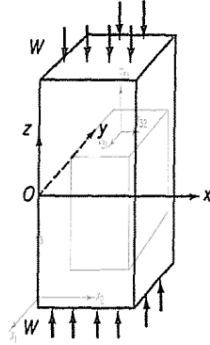
- Define: the dependent and independent variables of the mechanics of solids and fluid
- Define: Constitution, constitutional, constitutive

What is surprising, is the fact that three simple, idealized stress-strain relationships give a good description of the mechanical properties of many materials around us, namely, the nonviscous fluid, the Newtonian viscous fluid, and the perfectly elastic solid. We shall describe these idealized relations in this chapter, but we hasten to add that the properties of real materials differ more or less from the properties described by these idealized laws.

(§7.4) Hookean Elastic Solids

Example: Tensile Test

- How do materials respond to strain? or stress??



- $\epsilon_{33} \rightarrow \epsilon$, the normal strain. $\tau_{33} \rightarrow \sigma$, the normal stress.

$$\epsilon = \kappa \sigma, \quad \sigma = E \epsilon$$

- What is the *modulus of elasticity*, E , of the bar?

Generalization

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$$\begin{bmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{bmatrix} \quad ?? \quad \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{bmatrix}$$

- Connections between the stress and strain components? How many of them are there? C_{ijkl} ??
- Write out one component, e.g.,

$$\tau_{23} = C_{2311}\epsilon_{11} + C_{2312}\epsilon_{12} + C_{2313}\epsilon_{13} \dots + C_{2333}\epsilon_{33}$$

- There are eight other such components! $9 \times 9 = 81$ components in C_{ijkl} !!
- The general linear stress-strain relation is enclosed within a single indicial expression

$$\tau_{ij} = C_{ijkl}\epsilon_{kl}$$

Reduction

- The actual number of independent material parameters? 81? certainly not!

- **Important Note:** Stress is a symmetric tensor! Since $\tau_{ij} = \tau_{ji}$, we need

$$C_{ijkl}\epsilon_{kl} = C_{jikl}\epsilon_{kl} \quad \forall k, l$$

- This requires that $C_{ijkl} = C_{jikl}$, and is known as the first **minor symmetry** in the C tensor, reducing the number of coefficients, $81 \rightarrow 54$.
- **Another Note:** Strain is also symmetric!! By the freedom to arbitrarily assign dummy indices, we get

$$C_{ijkl}\epsilon_{kl} = C_{ijkl}\epsilon_{lk} = C_{ijlk}\epsilon_{kl}$$

- This means that $C_{ijkl} = C_{ijlk}$, and is known as the **second minor symmetry**, further reducing the number of coefficients, $54 \rightarrow 36$
- **Finally**, it can be shown that the coefficient tensor possesses **major symmetry** in the dual index pairs.

$$C_{ijkl} = C_{klij}$$

- Thus the most general elasticity tensor can be constructed in terms of a total of 21 elastic coefficients.

The Voight Solid

- We reshape the three-by-three stress and strain tensors as one-by-six first order arrays.

$$\begin{Bmatrix} \tau_{11} \\ \tau_{22} \\ \tau_{33} \\ \tau_{23} \\ \tau_{31} \\ \tau_{12} \end{Bmatrix} = \begin{bmatrix} C_{1111} & C_{1122} & C_{1133} & C_{1123} & C_{1113} & C_{1112} \\ & C_{2222} & C_{2233} & C_{2223} & C_{2213} & C_{2212} \\ & & C_{3333} & C_{3323} & C_{3313} & C_{3312} \\ & & & C_{2323} & C_{2313} & C_{2312} \\ \text{sym} & & & & C_{1313} & C_{1312} \\ & & & & & C_{1212} \end{bmatrix} \begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \epsilon_{23} \\ \epsilon_{31} \\ \epsilon_{12} \end{Bmatrix}$$

Isotropy

- Does material behavior change as a function of the stress traction vector?
- Are material properties a function of the orientation?

- If the answer is **no** or **almost no**, we call the material to be *isotropic*.
- In this case, the number of independent elasticity coefficients goes down, $21 \rightarrow 2$.
- For an isotropic material, exactly *two* independent elastic constants are required to characterize the material
- The constitutive relation for a Hookean solid is called the *Hooke's law* reading

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2\mu \epsilon_{ij}$$

where λ and μ are called *Lame's constants*.

in extenso

- It will be useful to expand the Hooke's law from indicial notation to an extensive form

$$\sigma_{xx} = \lambda(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}) + 2G\epsilon_{xx} \quad (1)$$

$$\sigma_{yy} = \lambda(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}) + 2G\epsilon_{yy} \quad (2)$$

$$\sigma_{zz} = \lambda(\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}) + 2G\epsilon_{zz} \quad (3)$$

$$\sigma_{yz} = G\epsilon_{yz}, \quad \sigma_{zx} = G\epsilon_{zx}, \quad \sigma_{xy} = G\epsilon_{xy}$$

where $G = \mu$ is a constant known as the *shear modulus*.

Inversion

- These equations can be inverted to write the strains in terms of stresses.
- It is customary to introduce **another pair** of material parameters in writing the inverted form of constitutive relations: the elastic modulus, E , and the Poisson's ratio, ν . These are related with λ and μ .
- The inverted form reads

$$\begin{aligned} e_{xx} &= \frac{1}{E}[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})], & e_{xy} &= \frac{1+\nu}{E} \sigma_{xy} = \frac{1}{2G} \sigma_{xy}, \\ e_{yy} &= \frac{1}{E}[\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})], & e_{yz} &= \frac{1+\nu}{E} \sigma_{yz} = \frac{1}{2G} \sigma_{yz}, \quad \blacktriangle (7.4-7) \\ e_{zz} &= \frac{1}{E}[\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})], & e_{zx} &= \frac{1+\nu}{E} \sigma_{zx} = \frac{1}{2G} \sigma_{zx}, \end{aligned}$$

Fluids

- The independent variable describing a fluid motion is the *velocity*.
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